Thanishkaa Saravanane

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Education

University of Washington, Seattle – Paul G. Allen School of Computer Science & Engineering
B.S. in Computer Science | Minors: Entrepreneurship, Neural Computation — *Expected Spring 2028*
Relevant Coursework: Data Structures and Parallelism, Hardware/Software Interface, Linear Algebra, Foundations of Computing, Software Design and Implementation, Intermediate Data Programming, Quantum Computing (Coursera), Cloud Computing and GCP (Coursera)

Experience

Research Intern — Jan 2026 – Present
Biorobotics Lab, University of Washington
- Engineered a real-time adaptive model to optimize driver lane-tracking performance using a game-theoretic Linear-Quadratic (LQ) framework and least-mean-squares (LMS) gradient descent to estimate human control policy from observed actions
- Built an experimental closed-loop lane-tracking control system (Python/Tkinter) with sum-of-sines disturbances, plant dynamics, and slider/Novint Falcon hardware inputs, collecting human control data at 60 Hz across 60+ trials
- Validated helper model using mean squared tracking error, human/helper policy gains, and FFT frequency analysis

Research Intern — Mar 2024 – Present
Bair Lab, University of Washington
- Recovered 15% of missed regions of interest (ROIs) across 1,000+ primate neurons by building a Python/Suite2p calcium image-processing pipeline on Linux and combining two image-segmentation methods via spatial overlap
- Computed 100+ neuronal tuning curves with SciPy to quantify signal quality (SNR, means, variances) and map cortical organization, revealing topographic clustering of neurons responsive to smaller visual targets
- Developed partial Vision Transformer (ViT) architectures with PyTorch/CUDA and computed Color Sensitivity Index across 196 patches and 12 encoder layers to compare ViT, AlexNet, and biological neuron responses

Software Engineer Intern — Jan 2026 – Jun 2026
Vaccine Genie
- Built a full-stack admin dashboard (React, JavaScript, MongoDB) with CRUD operations, input validation, and role-based access for 100+ management users
- Redesigned authentication architecture with a dedicated admin login flow and role-verified, token-based route protection to block non-admin access

Machine Learning Engineer Intern — Jul 2024 – Aug 2024
MulticoreWare
- Trained ResNet and U-Net models with PyTorch/TensorFlow in collaboration with ML teams for BMW and Qualcomm autonomous systems, optimizing performance with ONNX Runtime and Netron
- Built an NLP classification pipeline for large-scale spam-caller detection, achieving over 90% accuracy

Projects

Predictive Women's Health Engine | *Full-Stack ML and Two-Stage Clinical Risk Platform* — Apr 2026 – Present
- Training classifiers (XGBoost, Random Forest) across 10+ reproductive conditions (PCOS, Endometriosis, etc.) from 30+ patient symptoms, with current models achieving over 80% recall
- Integrating model predictions with reproductive history in a deterministic/RAG pipeline using 70+ literature-derived risk multipliers to project long-term cardiovascular risk in math-based reports for physician review of earlier preventive screening

Git Clarity | *VS Code Extension* — Apr 2026 – Present
- Engineering a developer tool that renders live branch histories as lane-based graphs via Git CLI, scans changes for exposed credentials, and uses Gemini API for spec-grounded analysis of each commit's codebase impact
- Building a Firebase Realtime Database that projects editable Gemini-assisted developer summaries onto live branch trees to flag merge-conflict risks via changed-file overlap

ML-Based EEG Classifier | *Full-Stack ML Inference Platform* — Apr 2024 – Jun 2025
- Trained and evaluated 5 models (Random Forest, Neural Network, RNN, ResNet, CNN-RNN) on 12,000+ EEG samples to predict student confusion during lectures, achieving 79.36% accuracy
- Preprocessed 36 EEG features and feature-engineered Theta/Beta and Theta/Alpha ratios as discriminative features
- Architected a scalable platform (React, FastAPI, WebSockets) to stream predictions across 5 concurrent models, with a real-time dashboard for accuracy, inference latency, and confusion matrices, plus a schema-validated API that generates dashboard analytics from user-uploaded EEG data to support BCI-driven personalized learning

Skills and Leadership

Languages: Java, Python, C, JavaScript, HTML
Machine Learning & Data: PyTorch, TensorFlow, XGBoost, scikit-learn, CUDA, ONNX Runtime, Pandas, NumPy, SciPy, Matplotlib, Suite2p, Retrieval-Augmented Generation (RAG), Computer Vision, Deep Learning, Model Optimization, Pearson Correlation, Fisher z-Transformation
Systems & Web: React, FastAPI, MongoDB, Firebase Realtime Database, WebSockets, Git, Force Dimension SDK, VS Code API, Gemini API, Linux, REST APIs, TypeScript, Rule-Based Systems, JUnit
Leadership: Founder of Smile Squad; raised $2K+ for Spirit of 12 Seahawks/NAMI; organized mental health support webinars and one-on-one personalized sessions for middle school students